

# Robustness & Deep Uncertainty

## Lecture

James Doss-Gollin

2026-03-02

### 1 Review

Over the past few weeks we built three tools. Each requires a probability distribution  $p(s)$  over states of the world.

**Benefit-cost analysis and policy search** (Labs 5–6). Choose the action  $a$  that maximizes expected net benefit:

$$a^* = \arg \max_a \sum_s U(a, s) p(s) \quad (1)$$

In Lab 5, you searched over dike heights to find the cost-minimizing action — that search is **policy search** (or simulation-optimization).

**Value of information** (Week 7). The expected value of perfect information is the gap between optimizing with and without knowledge of  $s$ :

$$\text{EVPI} = E_s \left[ \max_a U(a, s) \right] - \max_a E_s [U(a, s)] \quad (2)$$

EVII extends this to imperfect signals that update  $p(s)$  via Bayes' rule.

**Global sensitivity analysis** (Week 7 Wednesday). Sobol indices decompose output variance by input. The first-order index  $S_i$  measures the fraction of  $\text{Var} [Y]$  explained by  $X_i$  alone:

$$S_i = \frac{\text{Var}_{X_i} [E[Y | X_i]]}{\text{Var} [Y]} \quad (3)$$

#### 1.1 Laplace's Principle

What if we don't know  $p(s)$ ? A starting point: set  $p(s) \propto 1$  (equal weight on every scenario). This is **Laplace's principle of insufficient reason**:

$$a_{\text{Laplace}}^* = \arg \max_a \frac{1}{|S|} \sum_s U(a, s) \quad (4)$$

This is Equation 1 with a uniform prior. It is not assumption-free — it assumes every scenario is equally likely.

## 2 Deep Uncertainty

A spectrum of uncertainty:

- **Certainty** — single known future (Lab 5 under one RCP 8.5 trajectory)
- **Objective probabilities** — well-characterized distributions from data (GEV for storm surge, Labs 3–4)
- **Subjective probabilities** — expert-informed distributions (50/50 RCP prior in Lab 6)
- **Deep uncertainty** — experts disagree on which outcomes are even possible (SLR in 2150, tipping points, socioeconomic futures)

The boundary between subjective and deep uncertainty is fuzzy. The practical point: when  $p(s)$  is shaky, optimizing expected performance can give brittle answers.

R. J. Lempert and M. E. Schlesinger [1]: whoever controls the probability assumptions controls the policy conclusion. BCA's value is transparency, not correctness.

## 3 Robustness

N. C. Matalas and M. B. Fiering [2] defined robustness as:

“The insensitivity of system design to errors, random or otherwise, in the estimates of those parameters affecting design choice.”

A robust solution performs well not because we got the assumptions right, but because it doesn't depend too heavily on any particular assumption.

A **robustness metric** maps a performance matrix  $U(a, s)$  to a scalar score for each action, without requiring  $p(s)$ . The rest of this lecture develops several such metrics and compares what they recommend.

### 3.1 Notation and Case Study

We use a discrete example to build intuition, but the goal is to think about continuous action and scenario spaces.

- $A = \{a_1, \dots, a_m\}$ : available actions (dike heights, elevations, policies)
- $S = \{s_1, \dots, s_n\}$ : plausible scenarios (SLR pathways, climate states)
- $|S| = n$ : the number of scenarios (not absolute value)
- $U(a, s)$ : performance (net benefit) of action  $a$  under scenario  $s$
- $U$  with  $m$  actions and  $n$  scenarios is an  $m \times n$  **payoff matrix**

Three dike heights, three SLR scenarios. Entries are net benefit in \$M (avoided damages minus construction cost):

|                   | $s_1$ (Low SLR) | $s_2$ (Med SLR) | $s_3$ (High SLR) |
|-------------------|-----------------|-----------------|------------------|
| $a_1$ (Low dike)  | 9               | 2               | −6               |
| $a_2$ (Med dike)  | 3               | 4               | 1                |
| $a_3$ (High dike) | −2              | 3               | 8                |

## 4 Regret

Regret measures how much worse a solution performs compared to a reference point. J. D. Herman, P. M. Reed, H. B. Zeff, and G. W. Characklis [3] distinguish two types.

### 4.1 Baseline Regret

**Baseline regret** compares performance to the action's own average performance:

$$R_{\text{base}}(a, s) = \bar{U}(a) - U(a, s), \quad \bar{U}(a) = \frac{1}{|S|} \sum_s U(a, s) \quad (5)$$

Interpretation: how much worse did I do than my average? High baseline regret means this scenario was unusually bad for my chosen action — it's a measure of getting unlucky.

### 4.2 Opportunity-Cost Regret

**Opportunity-cost regret** (Savage regret) compares performance to the best possible action in that scenario:

$$R_{\text{opp}}(a, s) = \max_{a'} U(a', s) - U(a, s) \quad (6)$$

Interpretation: how much did I leave on the table?

Best action per scenario:  $\max_{a'} U(a', s_1) = 9$ ,  $\max_{a'} U(a', s_2) = 4$ ,  $\max_{a'} U(a', s_3) = 8$ .

The **regret matrix**:

|       | $s_1$           | $s_2$       | $s_3$           |
|-------|-----------------|-------------|-----------------|
| $a_1$ | $9 - 9 = 0$     | $4 - 2 = 2$ | $8 - (-6) = 14$ |
| $a_2$ | $9 - 3 = 6$     | $4 - 4 = 0$ | $8 - 1 = 7$     |
| $a_3$ | $9 - (-2) = 11$ | $4 - 3 = 1$ | $8 - 8 = 0$     |

Every action has zero regret in the scenario it's best for.

**Connection to VOI:** Opportunity-cost regret under perfect information is zero — you'd always pick the best action. EVPI (Equation 2) measures the expected value of eliminating this regret. Regret and value of information are two views of the same gap.

**Connection to BCA:** In BCA we chose the action maximizing expected net benefit. Minimax regret asks instead: which action minimizes the worst gap between what I got and what I could have gotten?

### 4.3 Minimax Regret

$$a_{\text{minimax}}^* = \arg \min_a \max_s R_{\text{opp}}(a, s) \quad (7)$$

Worst-case regret for each action:

- $a_1$ :  $\max(0, 2, 14) = 14$
- $a_2$ :  $\max(6, 0, 7) = 7$
- $a_3$ :  $\max(11, 1, 0) = 11$

$$a_{\text{minimax}}^* = a_2 \quad (\text{max regret} = 7)$$

## 4.4 Expected Regret

$$a_{\text{exp}}^* = \arg \min_a \sum_s w(s) R_{\text{opp}}(a, s) \quad (8)$$

The weights  $w(s)$  do not have to be equal. Equal weights ( $w(s) = 1/|S|$ ) are often assumed, but you can also set  $w(s) = p(s)$  — and then expected regret reduces to the expected opportunity cost of not having perfect information, which links directly back to BCA.

With equal weights:

- $a_1: (0 + 2 + 14)/3 = 5.3$
- $a_2: (6 + 0 + 7)/3 = 4.3$
- $a_3: (11 + 1 + 0)/3 = 4.0$

$$a_{\text{exp}}^* = a_3 \quad (\text{mean regret} = 4.0)$$

## 5 Satisficing

Regret asks “how far am I from the best?” A different family asks “is my performance good enough?” — these are **satisficing** metrics [4].

### 5.1 Domain Criterion

The **domain criterion** measures the fraction of scenarios in which a solution meets a performance threshold  $\tau$ :

$$D(a) = \frac{1}{|S|} \sum_{s \in S} \mathbf{1}[U(a, s) \geq \tau] \quad (9)$$

where  $|S| = n$  is the number of scenarios.

This is meaningful when hard thresholds exist in practice:

- A levee must not be overtopped
- A budget must not be exceeded
- A species population must not drop below a viability threshold
- A road must remain passable during a design storm

The domain criterion counts: over what fraction of futures does my solution avoid failure?

Set  $\tau = 1$  (net benefit at least \$1M):

- $a_1: U = 9 \geq 1 \checkmark, U = 2 \geq 1 \checkmark, U = -6 < 1 \times \rightarrow D(a_1) = 2/3$
- $a_2: U = 3 \geq 1 \checkmark, U = 4 \geq 1 \checkmark, U = 1 \geq 1 \checkmark \rightarrow D(a_2) = 3/3 = 1.0$
- $a_3: U = -2 < 1 \times, U = 3 \geq 1 \checkmark, U = 8 \geq 1 \checkmark \rightarrow D(a_3) = 2/3$

$$a_{\text{domain}}^* = a_2 \quad (D = 1.0)$$

The result depends on  $\tau$ . If  $\tau = 4$ : three-way tie. If  $\tau = 3$ : tie between  $a_2$  and  $a_3$ . The threshold is a value judgment.

## 5.2 Maximin

$$a_{\text{maximin}}^* = \arg \max_a \min_s U(a, s) \quad (10)$$

Worst-case performance:

- $a_1$ :  $\min(9, 2, -6) = -6$
- $a_2$ :  $\min(3, 4, 1) = 1$
- $a_3$ :  $\min(-2, 3, 8) = -2$

$$a_{\text{maximin}}^* = a_2 \quad (\text{worst case} = 1)$$

Maximin on payoffs and minimax on regrets are related: both focus on worst-case protection. Maximin asks “what’s the worst that can happen to me?” Minimax regret asks “what’s the worst gap between my choice and the best I could have made?”

## 5.3 Maximax

$$a_{\text{maximax}}^* = \arg \max_a \max_s U(a, s) \quad (11)$$

Best-case performance:

- $a_1$ :  $\max(9, 2, -6) = 9$
- $a_2$ :  $\max(3, 4, 1) = 4$
- $a_3$ :  $\max(-2, 3, 8) = 8$

$$a_{\text{maximax}}^* = a_1 \quad (\text{best case} = 9)$$

## 6 Synthesis

Six metrics, same payoff matrix, three different winners:

| Metric                | Winner | What it measures                 |
|-----------------------|--------|----------------------------------|
| Minimax regret        | $a_2$  | Worst-case regret = 7            |
| Expected regret       | $a_3$  | Mean regret = 4.0                |
| Domain ( $\tau = 1$ ) | $a_2$  | Fraction meeting threshold = 1.0 |
| Maximin               | $a_2$  | Worst-case payoff = 1            |
| Maximax               | $a_1$  | Best-case payoff = 9             |
| Laplace               | $a_3$  | Mean payoff = 3.0                |

Same physical system, same scenarios, same payoffs — the answer changed because the *question* changed.

The structural driver is **performance crossovers** [4].  $a_1$  outperforms  $a_3$  under low SLR (9 vs.  $-2$ ), but  $a_3$  outperforms  $a_1$  under high SLR (8 vs.  $-6$ ). When performance curves cross, the metric’s

risk attitude breaks the tie. If one action dominated another across all scenarios, every metric would agree.

The choice of metric embeds assumptions about:

- What “good enough” means ( $\tau$  in the domain criterion)
- How much you care about worst cases vs. averages (minimax vs. Laplace)
- What the reference point is (regret from baseline vs. regret from best)

These choices should be made explicitly and transparently.

## 6.1 Robustness and Probability

Write the domain criterion in continuous form. In general, the satisficing constraint may be on any performance metric  $g(a, s)$  (cost, damage, reliability, etc.), not just raw utility:

$$D(a) = \int \mathbf{1}[g(a, s) \geq \tau] p(s) ds \quad (12)$$

where  $p(s)$  is the sampling density over scenarios. In our case study  $g = U$ , but in multi-objective problems we might have separate thresholds on cost, flood depth, ecosystem impact, etc.

- Sample 1000 SLR scenarios uniformly on  $[0, 2]$  meters:  $D = 0.70$
- Sample 1000 scenarios concentrated around  $[0.3, 0.7]$  meters:  $D = 0.90$

Same dike, same threshold, same physics — different score. The sampling distribution  $p(s)$  changed the answer.

Similarly, Laplace’s criterion (Equation 4) computes mean performance with equal weights — identical to expected utility under a uniform prior.

Robustness metrics make implicit distributional assumptions through their choice of which scenarios to include and how to weight them. The claim that robustness avoids probabilities is, at best, incomplete.

A related warning from N. C. Matalas and M. B. Fiering [2]: a design yielding payoff 2 in every scenario scores well on maximin ( $\min = 2$ ) and the domain criterion ( $D = 1.0$  for  $\tau \leq 2$ ), but is never optimal under any scenario. Robustness is a property you want your solution to have, not the only property you care about.

## 6.2 Looking Ahead

**Wednesday:** The Lempert-Schneider debate. R. J. Lempert and M. E. Schlesinger [1] argues that prediction-based policy analysis breaks down under deep uncertainty and that we should search for robust strategies using exploratory modeling. S. H. Schneider [5] argues that risk management *requires* probabilities — without them, we can’t prioritize. Both papers are in this week’s reading guide. Come ready to take sides.

One constructive response: J. Doss-Gollin and K. Keller [6] propose making distributional assumptions explicit, computing performance under multiple probability distributions, and calling a solution robust only if it holds up across different distributional assumptions — not just across different parameter values.

## 6.3 Final Project

Teams and plans are assigned today. Key deadlines:

| Deadline                       | Deliverable                                                     |
|--------------------------------|-----------------------------------------------------------------|
| <b>Week 10</b> (Fri., Mar. 27) | Memo 1: XLRM Framework (Uncertainties, Levers, Models, Metrics) |
| <b>Week 12</b> (Fri., Apr. 10) | Memo 2: Evidence & Robustness (Valuation & Deep Uncertainty)    |
| <b>Week 13</b> (Fri., Apr. 17) | Slides Due                                                      |
| <b>Week 14</b>                 | Executive Briefing (Presentations)                              |
| <b>Finals</b>                  | Written Report                                                  |

See Final Project for full details.

### 6.3.1 Graduate Seminar (CEVE 521)

Graduate students lead one Wednesday paper discussion session in small groups. Paper approval is due **2 weeks before** your session.

| Group   | Session                 | Topic                | Paper Approval By | Materials Posted By |
|---------|-------------------------|----------------------|-------------------|---------------------|
| Group 1 | Wed., Mar. 25 (Week 10) | Deep Uncertainty     | Mar. 11           | Mar. 18             |
| Group 2 | Wed., Apr. 1 (Week 11)  | Exploratory Modeling | Mar. 18           | Mar. 25             |
| Group 3 | Wed., Apr. 8 (Week 12)  | Sequential Decisions | Mar. 25           | Apr. 1              |

See Graduate Seminar Requirements for full details.

## 7 References

### Bibliography

- [1] R. J. Lempert and M. E. Schlesinger, "Robust Strategies for Abating Climate Change," *Climatic Change*, vol. 45, no. 3–4, pp. 387–401, Jun. 2000, doi: 10.1023/A:1005698407365.
- [2] N. C. Matalas and M. B. Fiering, "6. Water-Resource Systems Planning,," *Climate, Climatic Change, and Water Supply*. The National Academies Press, Washington, DC, pp. 99–110, 1977. [Online]. Available: <https://www.nap.edu/read/185/chapter/11>
- [3] J. D. Herman, P. M. Reed, H. B. Zeff, and G. W. Characklis, "How Should Robustness Be Defined for Water Systems Planning under Change?," *Journal of Water Resources Planning and Management*, vol. 141, no. 10, p. 4015012, Oct. 2015, doi: 10.1061/(asce)wr.1943-5452.0000509.
- [4] C. McPhail, H. R. Maier, J. H. Kwakkel, M. Giuliani, A. Castelletti, and S. Westra, "Robustness Metrics: How Are They Calculated, When Should They Be Used and Why Do They Give Different Results?," *Earth's Future*, pp. 169–191, Apr. 2019, doi: 10.1002/2017ef000649.

- [5] S. H. Schneider, "Can We Estimate the Likelihood of Climatic Changes at 2100?," *Climatic Change*, vol. 52, no. 4, pp. 441–451, Mar. 2002, doi: <http://dx.doi.org/10.1023/A:1014276210717>.
- [6] J. Doss-Gollin and K. Keller, "A Subjective Bayesian Framework for Synthesizing Deep Uncertainties in Climate Risk Management," *Earth's Future*, vol. 11, no. 1, Jan. 2023, doi: [10.1029/2022EF003044](https://doi.org/10.1029/2022EF003044).